TitrateRL: A Simulation-Based Reinforcement Learning Approach for Personalized Basal Insulin Titration in Type 2 Diabetes

Boburmirzo Uchqunov
ID 240447
Software engineering, CI, dashboard

Nodirjon Khabibullaev
ID 230019
Literature, evaluation, statistics

Ibrohim Abralov
ID 230129
Simulator, cohort, validation

Abdurashid Abdusattarov
ID 221229
RL agents, rewards, safety shield

School of Engineering, BSc in Computer Science, Central Asian University
Team TitrateRL Supervisor: Dr. Eugene Castro November 2026

***Abstract*—Basal insulin in Type 2 diabetes (T2D) is titrated with population-level rules that apply the same fasting-glucose thresholds to every patient, and no open, reinforcement-learning-ready environment exists to test whether learned policies can do better safely. This paper presents TitrateRL, an open-source framework that combines a calibrated virtual T2D cohort, a Gymnasium environment for daily basal-insulin titration, Proximal Policy Optimization agents with a rule-based safety shield, and a pre-specified evaluation against three clinical titration algorithms. The methods section describes the system architecture, deployment, repository structure, class design and the minimum viable product defined with the MoSCoW method. The software is implemented and tested; the experimental results are reported in the final version. All work is in silico and not intended for clinical use. Code: https://github.com/Nodirjn/TitrateRL.**

***Index Terms*—reinforcement learning, Type 2 diabetes, basal insulin titration, in silico simulation, safe RL, Gymnasium**

I. Introduction

A. Background

An estimated 589 million adults lived with diabetes in 2024, and over 90% of them have Type 2 diabetes (T2D) [1]. When non-insulin agents no longer reach the glycemic target, a once-daily long-acting basal insulin such as insulin glargine is commonly added. Its dose is titrated: guidelines recommend starting at 10 U/day and increasing the dose, for example by 2 U every three days, until the fasting plasma glucose (FPG) reaches 80–130 mg/dL [2], [3]. Such algorithms were validated in large trials [4]–[6], but in routine care up-titration is slow: in a large real-world cohort, 81% of patients remained above their HbA1c target six months after starting basal insulin [7]. Continuous glucose monitoring (CGM) now provides consensus metrics such as time in range (TIR, 70–180 mg/dL) [8], and physiological simulators are accepted for preclinical testing of insulin-dosing algorithms [9].

B. Problem and Motivation

Clinical titration algorithms apply one fasting-glucose rule to every patient, although insulin needs differ widely and lower body-mass index and lower doses predict hypoglycemia [10]. Reinforcement learning (RL) suits sequential dosing decisions with delayed effects [11], but of 29 RL glucose-control studies in a systematic review, 24 addressed Type 1 diabetes and only 2 addressed T2D alone [12]. Existing T2D approaches use non-public hospital data [13] or commercial simulators [14], and open simulators either contain only Type 1 patients [15] or keep the basal dose fixed [16]. Hidden insulin sensitivity, delayed effects and asymmetric risk make daily titration a safety-constrained partially observable Markov decision process (POMDP).

C. Research Questions and Objectives

The general objective is to design, implement and evaluate within 14 weeks an open-source, simulation-based RL framework for daily basal-insulin titration in virtual patients with T2D, benchmarked against clinical titration algorithms with consensus glycemic metrics.

- **RQ1 / SO1:** How does the reward formulation (R1 time in range, R2 risk index, R3 time in range with hypoglycemia penalties) change the trade-off between time in range and time below range? *H1: R3 lowers $TBR_{<70}$ at comparable TIR.*
- **RQ2 / SO2:** Does a shielded PPO policy achieve a higher maintenance-phase TIR than the AT.LANTUS, Treat-to-Target and INSIGHT algorithms on unseen patients while remaining non-inferior in time below range? *H2: gain $\geq$5 pp, non-inferiority margin 1 pp.*
- **RQ3 / SO3:** Does the benefit depend on insulin sensitivity, and does one policy personalize its dosing? *H3: negative correlation with insulin sensitivity.*

II. Methods

A. Development Methodology

The project follows an iterative, Scrum-inspired process with two-week sprints, a backlog in GitHub Projects and supervisor reviews at milestones M1–M4. The simulator, wrappers and metrics are developed test-first with `pytest`;

GitHub Flow with a protected `main` branch, reviewed pull requests and continuous integration (ruff linting and tests on Python 3.10–3.12) is used. Every experiment is a version-controlled YAML file with fixed seeds, and results are stored with the commit hash that produced them.

B. System Architecture and Deployment

TitrateRL follows a layered architecture in which each layer depends only on the layers below it: configuration and data, simulation core, environment, decision, experiment and evaluation, and presentation. Fig. 1 shows where each part runs. Code flows from the team's workstations through pull requests and CI to tagged releases; experiments run on a multi-core CPU machine from a pinned commit; trained models and result tables are published as release assets and consumed by the dashboard and by researchers who install the package. The system has no server back end and handles no personal data.

C. Repository Structure

The repository (Fig. 2) follows the `src` layout and mirrors the architecture layers. Generated outputs (`data/`, `runs/`, `results/`, `reports/`) are git-ignored; milestones are tagged `v0.1-env`, `v0.2-agents`, `v0.3-eval` and `v1.0-mvp`.

D. Class Design

Fig. 3 shows the main classes. `T2DTitrationEnv` composes a `VirtualPatient`, which owns the parameter vector, the ODE state and three random streams (meals, CGM, fingerstick) so that all policies can be evaluated with common random numbers. Rule-based baselines and RL agents share one abstract `Policy` interface; `SafetyShield` is a Gymnasium wrapper applied only to RL agents, whereas the rate limit and dose bounds apply to every policy.

E. Virtual Patient Model and Cohort

The patient is described by the meal simulation model of Dalla Man *et al.* with its published T2D parameters [17], [18]. The glucose, insulin and gut subsystems are adapted from the open-source `simglucose` implementation [15] and extended with β-cell secretion, time-varying hepatic insulin extraction and a two-compartment subcutaneous glargine model [19], [20]. The 15-state system is integrated with a fixed-step fourth-order Runge–Kutta scheme (1-min step) compiled with Numba [21]. Each simulated day contains three meals with random timing and size and an optional snack; glucose is observed by a CGM every 5 min with the noise model of Breton and Kovatchev [22] and by a fasting fingerstick at 07:00 with ISO 15197:2013-compatible error [23]. A cohort of 150 virtual adults with varied pre-insulin FPG, body weight, insulin sensitivity (ρ_{SI}) and β-cell function (ρ_β) is calibrated against the Treat-to-Target trial [4] and split 90/30/30 into training, validation and test patients, stratified by ρ_{SI} tertile.

F. Decision Process and Rewards

Titration is formulated as a POMDP exposed through the Gymnasium API [24]. One step is one day and one episode lasts 168 days (24 weeks). The observation stacks 11 daily features (fasting SMBG, weight-normalized dose, last dose change, CGM TIR, TBR, TAR, mean and coefficient of variation, recent hypoglycemic events, body weight and time) over the last three days. The action is a dose change in $\{-4, -2, 0, +2, +4\}$ U; for every policy the cumulative increase over any seven days is limited to 8 U and the dose to $[0, 1.0$ U/kg$]$. Three daily rewards are compared: R1 $=$ TIR; R2 $= -\frac{1}{K}\sum_k \rho(G_k)/100$, with the symmetrized risk function of Kovatchev *et al.* [25]; and R3 $=$ TIR $- 5\,\mathrm{TBR}_{<70} - 10\,\mathrm{TBR}_{<54}$.

G. Agents, Safety Shield and Baselines

PPO [26] is the primary algorithm and DQN [27] a secondary one, both from Stable-Baselines3 [28]; each configuration is trained for 10^6 simulated days with five seeds [29], a new random training patient every episode, and checkpoint selection on validation patients only. A rule-based safety shield inspired by shielding in safe RL [30] blocks dose increases after level-1 hypoglycemia in the last 72 h and reduces the dose by 20% after level-2 hypoglycemia in the last 24 h. The baselines are a fixed dose of 0.2 U/kg (B0), AT.LANTUS (B1) [5], Treat-to-Target (B2) [4], INSIGHT (B3) [6] and a simulator-tuned threshold rule (B4).

H. Evaluation Protocol and Metrics

Every policy is evaluated on the 30 test patients, each under ten 168-day scenarios with common random numbers. Endpoints are computed from noise-free plasma glucose (Table I) following the international CGM consensus [8], [31]. The primary efficacy endpoint is TIR in weeks 21–24 and the primary safety endpoint is $\mathrm{TBR}_{<70}$ over all 24 weeks. H2 is tested with Holm-adjusted one-sided paired Wilcoxon tests, the Hodges–Lehmann estimate of the TIR gain and 95% bootstrap confidence intervals; H3 with Spearman's correlation. Overbasalization signals [32] and the glucose management indicator [33] are reported as secondary outcomes.

I. Minimum Viable Product (MoSCoW)

The MVP is the minimal set of working features that demonstrates the complete pipeline: virtual patients, training and shielded titration, statistical comparison and interactive visualization. Table II lists the features prioritized with the MoSCoW method and their acceptance criteria.

J. Ethical Considerations

Only simulated patients are used; no human subjects, personal data or clinical records are involved. TitrateRL will not recommend doses for real people, is not a medical device, and labels every output as a simulation result, following guidelines for RL in healthcare [34].

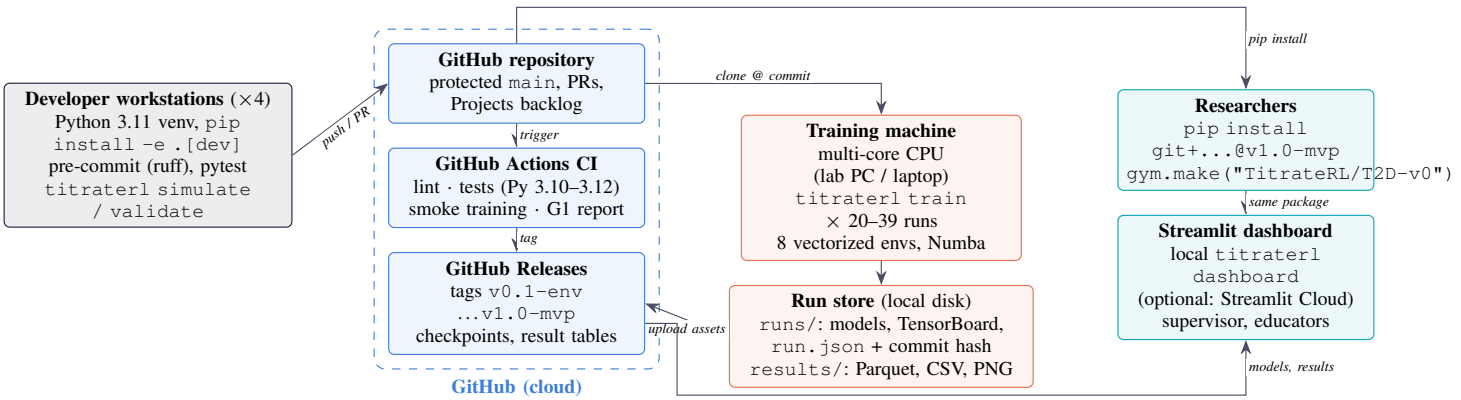


Fig. 1. Deployment map. Code flows from the four workstations through pull requests and CI to tagged releases; experiments run on a multi-core CPU machine from a pinned commit; trained models and result tables are published as release assets (never committed to git) and consumed by the dashboard and by researchers who install the package.

```
titraterl/
├─ .github/  — CI workflow, PR and issue templates
├─ configs/
│  ├─ cohort.yaml
│  ├─ train/  — ppo_r1/r2/r3, ppo_rstar_shield, dqn, fpg, smoke
│  └─ eval/  — test.yaml, baselines_quick.yaml
├─ src/titraterl/
│  ├─ simulation/  — parameters, model (ODE, RK4), scenario, sensors, patient
│  ├─ cohort/  — generator, split, Parquet I/O
│  ├─ envs/  — titration_env, observation, rewards, wrappers
│  ├─ policies/  — base, rules (B0–B4), sb3
│  ├─ training/  — train, callbacks
│  ├─ metrics/  — glycemic, episode
│  ├─ evaluation/  — runner, statistics, logging, pipeline
│  ├─ validation/  — checks V1–V5
│  └─ config.py, cli.py
├─ dashboard/app.py
├─ tests/
│  ├─ unit/  — model, cross-impl., env, shield, rules, ...
│  └─ integration/  — pipeline, CLI, training, dashboard
├─ docs/  — architecture, model, environment, evaluation, ...
├─ examples/  — quickstart.py, quickstart.ipynb
└─ README.md, MODEL_CARD.md, pyproject.toml, Makefile, LICENSE
```

Fig. 2. Repository file structure (86 tracked files, ≈3 800 lines of package code and ≈800 lines of tests).

TABLE I
EVALUATION AND ACCEPTANCE METRICS

Metric	Definition	Target / role
Product (acceptance)		
Test suite	unit + integration tests in CI	100% pass
Episode runtime	one 168-day episode	<2 s
Dashboard re-run	24-week simulation	<5 s
Simulator checks	V1–V4 (gate G1)	all pass by Wk 5
Reproducibility	same seed $\Rightarrow$ same trajectory	exact
Research (glycemic)		
TIR	% time 70–180 mg/dL, weeks 21–24	$>70\%$; primary efficacy
$\text{TBR}_{<70}$, $\text{TBR}_{<54}$	% time below 70 / 54 mg/dL	$<4\%$, $<1\%$; primary safety
$\text{TAR}_{>180}$, $\text{TAR}_{>250}$	% time above 180 / 250	$<25\%$, $<5\%$
Mean, CV, GMI	mean, SD/mean, $3.31+0.02392\,\bar{G}$	CV $\lesssim 36\%$
LBGI, HBGI	Kovatchev risk indices	risk
Time-to-target	3 consecutive fasting values 80–130	titration speed
Nocturnal TBR	00:00–06:00	safety
Overbasalization	bedtime−morning ≥ 50; dose >0.5 U/kg	appropriateness
Dose burden	final U/kg; changes per week	burden
Shield rate	% days with intervention	safety
Statistical decision rules		
H2 superiority	Holm-adj. one-sided Wilcoxon	$p < 0.05$
H2 effect size	Hodges–Lehmann TIR gain	≥ 5 pp
H2 non-inferiority	upper 95% CI of $\text{TBR}_{<70}$ diff.	<1 pp
H3	Spearman(ρ_{SI}, ΔTIR)	negative, $p < 0.05$

III. Results

A. Overall Framework Technical Evaluation (General Objective)

[Insert Results for General Objective here: end-to-end pipeline status, simulator verification and validation checks (gate G1), runtime per episode, training throughput and test-suite results.]

B. Reward Design and Efficacy–Safety Trade-off (SO1)

[Insert Results for SO1 here: TIR, $\text{TBR}_{<70}$ and $\text{TBR}_{<54}$ of PPO with R1, R2 and R3 on the test cohort (median and 95% bootstrap CI), share of patients meeting the consensus targets, and the H1 test.]

C. Comparison with Clinical Titration Algorithms (SO2)

[Insert Results for SO2 here: maintenance-phase TIR and full-period $\text{TBR}_{<70}$ of the shielded PPO policy versus B1–B3, Hodges–Lehmann gains, Holm-adjusted p-values, non-inferiority intervals, time-to-target and dose burden.]

D. Personalization and Insulin Sensitivity (SO3)

[Insert Results for SO3 here: Spearman correlation between ρ_{SI} and the TIR benefit, dose changes at matched fasting glucose by insulin-sensitivity tertile, and the train–test generalization gap.]

IV. Discussion

A. Overall Framework Evaluation and Technical Feasibility

[Insert Discussion for General Objective here: interpretation of simulator validity and calibration deviations, reproducibility of the pipeline, and project scope boundaries.]

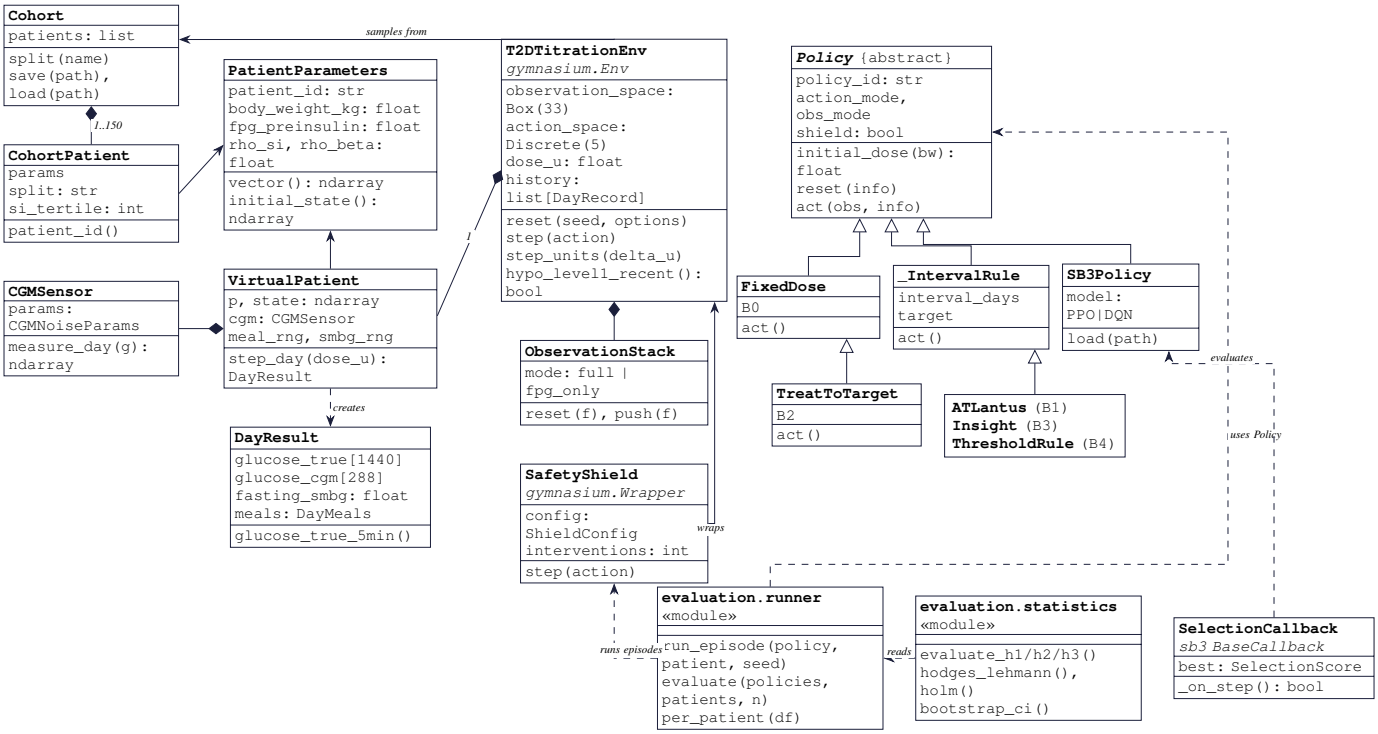


Fig. 3. Class diagram of the implemented TitrateRL package (simplified UML). Open triangles: inheritance; filled diamonds: composition; solid arrows: association; dashed arrows: dependency. B1, B3 and B4 inherit from `_IntervalRule`; B2 from `Policy`.

TABLE II
MVP PRIORITIZATION WITH THE MOSCOW METHOD

Priority	Feature	Acceptance criterion	Status
Must	F1 T2D simulator and calibrated 150-patient cohort	generated reproducibly from one seed; checks V1–V4 pass	implemented; V4 open
Must	F2 Gymnasium environment `TitrateRL/T2D-v0`	passes `check_env`; 168-day episode <2 s; rate-limit and shield tests pass	done (0.15 s)
Must	F3 Trained agents: PPO with R1–R3 (shield off) and shielded PPO with $R^\star$, 5 seeds each	checkpoints and learning curves stored; selection on validation patients	pipeline done; runs Wk 7–10
Must	F4 Rule-based policies B0–B3 behind the `Policy` interface	unit tests reproduce the published titration steps	done
Must	F5 Evaluation pipeline	one command regenerates metric tables, statistical tests and figures	done
Must	F6 Interactive dashboard	select a test patient and ≤ 3 policies; traces, doses, metrics, disclaimer; 24-week re-run <5 s; CSV export	done
Must	F7 Installable package	`pip install`; environment registered; documented API and example notebook	done
Must	RQ1–RQ3 analyses	H1–H3 decision rules applied to test-cohort data	Wk 10–13
Should	DQN agent; FPG-only PPO; λ grid for R3; simulator-tuned rule B4; robustness scenario set	completed and reported as exploratory	code ready
Could	Oracle per-patient fine-tuning; 2013 hypoglycemia model extensions; LSTM policy; SAC	started only if M3 is on time	—
Won't (this semester)	Real patient data or clinical deployment; Type 1 diabetes, pumps and bolus insulin; exercise, illness, missed injections; offline RL from EHRs; mobile app	out of scope; stated in README and model card	—

B. Reward Design (RQ1 / SO1)

[Insert Discussion for RQ1 here: interpretation of the efficacy–safety trade-off of the rewards, overbasalization risk, and implications for encoding asymmetric clinical objectives (Problem 1).]

C. Learned versus Clinical Titration (RQ2 / SO2)

[Insert Discussion for RQ2 here: interpretation of the comparison with B1–B3, the value of CGM features versus fasting-glucose-only observations, decision frequency, and simulator-specific tuning (Problem 2).]

D. Personalization (RQ3 / SO3)

[Insert Discussion for RQ3 here: whether the policy adapts to unobserved insulin sensitivity, generalization to unseen patients, and safety under heterogeneity (Problem 3).]

E. Limitations

[Insert Limitations here: absence of counter-regulation in the model, extrapolation of single-meal parameters to 24 weeks, perfect adherence, remaining calibration gap, and the simulation-to-reality gap.]

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